Analysing Severe Adverse Effects

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Contents

data wrangling

```
sae_df <- read_xlsx("data/Q2b.xlsx") |>
  janitor::clean_names() |>
  mutate(sae = ifelse(sae == 1, "Yes", "No") |> as.factor(),
         group = ifelse(group == 1, "Vaccine", "Control") |> as.factor())
#time = as.factor(time),


bl_df <- read_xlsx("data/Q2b_BL.xlsx") |>
  janitor::clean_names() |>
  mutate(sex = ifelse(sex == 1, "Male", "Female"),
         sex = fct_relevel(sex, "Male"))

combined_dataset <- sae_df |>
  full_join(bl_df, by = "id")

combined_dataset_wide <- combined_dataset |>
  pivot_wider(names_from = "time",
              values_from = "sae",
              names_prefix = "month_")
```

Summary Table

```
table1  <- combined_dataset %>%
  drop_na() |>
  dplyr::select(time:age) %>%
  tbl_summary(
    by = group,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      time ~ 'Time (months)',
      age ~ "Age (years)",
      sex ~ "Sex",
```

```
      sae ~ "Severe Adverse Events (Yes)",
      site ~ "Number of Sites")
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels() %>%
  modify_caption("**Baseline Characteristics by Treatment Group**")
# table1v2 <- table1 %>%
#   as_gt() %>%
#   gt::tab_options(
#     data_row.padding = px(2),
#     table.width      = pct(100),
#     table.border.top.color = "black",
#     table.border.bottom.color = "black",
#     heading.border.bottom.color = "black",
#     table.font.color = "black",
#     column_labels.border.bottom.style = "none", # Remove header line
#     table_body.border.bottom.style = "none",    # Remove body line
#     table_body.border.top.style = "none",
#     table.border.top.style = "none",
#     table.font.names = "arial",
#     table.font.size = "12px"
#   ) |>
#   cols_width(
#     label      ~ px(150),
#     everything() ~ px(80)
#   ) |>
#   gt::tab_style(
#     style = gt::cell_borders(),
#     locations = gt::cells_body(columns = everything()))  |>
#   gt::gtsave(filename = "figures_tables/summarytab.pdf")
#system("pdfcrop figures_tables/summarytab.pdf figures_tables/summarytab.pdf")

table1 |>
  as_gt() |>
  cols_width(
    label      ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::gtsave(filename = "figures_tables/summarytab.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmpkXafD0/file16d3272bc0559.html screensh
```

```
table1 |>
  as_kable()
```

Table 1: **Baseline Characteristics by Treatment Group**

Characteristic	**Overall** N = 111,452	**Control** N = 58,781	**Vaccine** N = 52,671	**p-value**
Time (months)				<0.001

Characteristic	Overall N = 111,452	Control N = 58,781	Vaccine N = 52,671	p-value
1	38,588 (35%)	20,014 (34%)	18,574 (35%)	
2	37,212 (33%)	19,608 (33%)	17,604 (33%)	
3	35,652 (32%)	19,159 (33%)	16,493 (31%)	
Severe Adverse Events (Yes)	175 (0.2%)	80 (0.1%)	95 (0.2%)	0.062
Number of Sites	50 (29)	50 (29)	50 (29)	0.7
Sex				0.015
Male	56,076 (50%)	29,373 (50%)	26,703 (51%)	
Female	55,376 (50%)	29,408 (50%)	25,968 (49%)	
Age (years)	45 (11)	45 (10)	45 (12)	0.060

Checking for missing data pattern (if any)

```
fig2a <- vis_miss(combined_dataset_wide) +
  theme(axis.text.x = element_text(vjust = 0))
ggsave("figures_tables/missingdata_info.png", fig2a, height = 5, width =8)
```

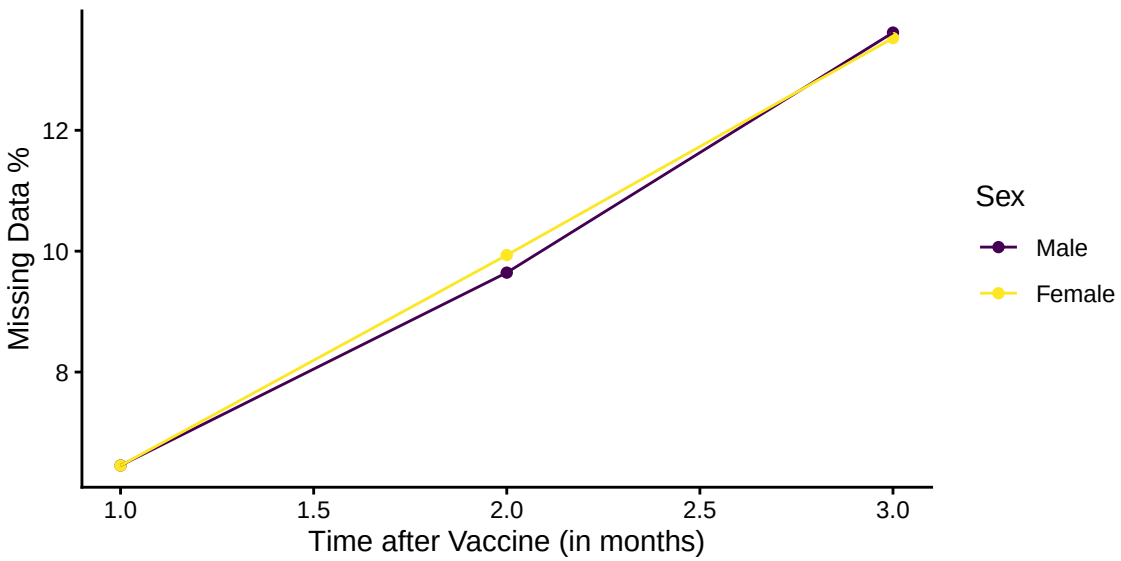
Missingness only seen in outcome data (SAE) and more so for 2nd and third timepoints

Missingness by gender

```
tx_gender_missing <- combined_dataset |>
  group_by(sex, time) |>
  mutate(total = length(unique(id))) |>
  ungroup() |>
  drop_na() |>
  group_by(sex, time) |>
  summarize (missing_pct = (total[1] - length(unique(id)))*100/total[1])
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by sex and time.
## i Output is grouped by sex.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(sex, time))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
missing_by_gender <- ggplot(aes(x = time, y = missing_pct, group = sex, color = sex),
       data = tx_gender_missing) +
  geom_point() +
  scale_x_continuous(limits = c(1, 3))+
  geom_line() +
  labs(x = "Time after Vaccine (in months)", y = "Missing Data %", color = "Sex") +
  theme_classic()

ggsave("figures_tables/missingness_by_gender.pdf", missing_by_gender, height = 3, width =5)
#system("pdfcrop figures_tables/missingness_by_gender.pdf figures_tables/missingness_by_gender.pdf")
missing_by_gender
```



Characterise missingness pattern

Number of drop outs

```
missingpattern <- combined_dataset_wide|>
  mutate(pattern = case_when(is.na(month_3) == FALSE ~ "Complete",
                             is.na(month_3) == TRUE & is.na(month_2) == TRUE &
                               is.na(month_1) == TRUE ~ "Assessment Dropout",
                             is.na(month_3) == TRUE & is.na(month_2) == TRUE &
                               is.na(month_1) == FALSE ~ "Early Dropout",
                             is.na(month_3) == TRUE ~ "Late Dropout"),
         pattern = fct_relevel(pattern, "Assessment Dropout", "Early Dropout",
                               "Late Dropout"))
missingpattern1 <- missingpattern |>
  dplyr::select(c(sex, pattern)) |>
  tbl_summary(
    by = pattern,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      sex ~ "Sex")
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels()

missingpattern1 |>
  as_gt() |>
  gt::gtsave(filename = "figures_tables/missingpattern1.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmpkXafD0/file16d324c134f79.html screensho
```

```
missingpattern1 |>
  as_kable()
```

Characteristic	Overall N = 41,250	Assessment Dropout N = 56	Early Dropout N = 605	Late Dropout N = 4,937	Complete N = 35,652	p-value
Sex						0.4
Male	20,747 (50%)	25 (45%)	288 (48%)	2,512 (51%)	17,922 (50%)	
Female	20,503 (50%)	31 (55%)	317 (52%)	2,425 (49%)	17,730 (50%)	

```
missingpatternv2 <- missingpattern |>
  dplyr::select(c(group, pattern)) |>
  tbl_summary(
    by = group,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      pattern ~ "Missingness Pattern")
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels()

missingpatternv2 |>
  as_gt() |>
  gt::gtsave(filename = "figures_tables/missingpatternv2.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmpkXafD0/file16d325ebe4f82.html screensho
```

```
missingpatternv2 |>
  as_kable()
```

Characteristic	Overall N = 41,250	Control N = 20,625	Vaccine N = 20,625	p-value
Missingness Pattern				<0.001
Assessment Dropout	56 (0.1%)	2 (<0.1%)	54 (0.3%)	
Early Dropout	605 (1.5%)	59 (0.3%)	546 (2.6%)	
Late Dropout	4,937 (12%)	1,405 (6.8%)	3,532 (17%)	
Complete	35,652 (86%)	19,159 (93%)	16,493 (80%)	

Only 3.7% missingness, ignoring for now

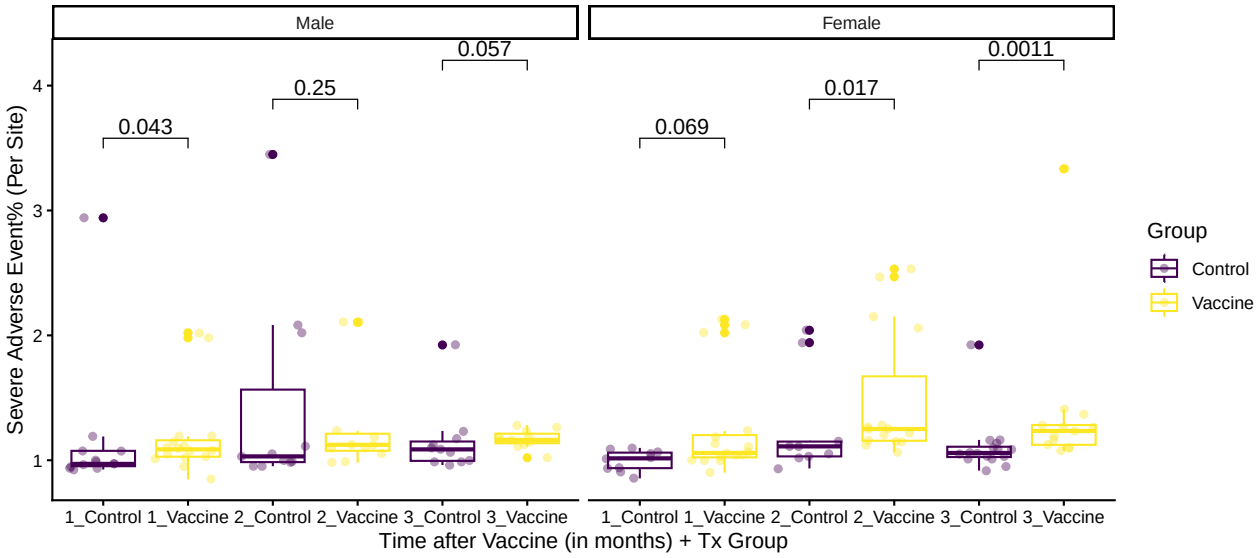
EDA for SAE

```
sae_pct <- combined_dataset |>
  drop_na() |>
  group_by(group, time, site, sex) |>
```

```
  mutate(total = n()) |>
  ungroup() |>
  dplyr::filter(sae == "Yes") |>
  group_by(group, time, site, sex) |>
  summarise (sae_pct = n()*100/total[1]) |>
  mutate(group_time = paste(time, group, sep = "_"))
```

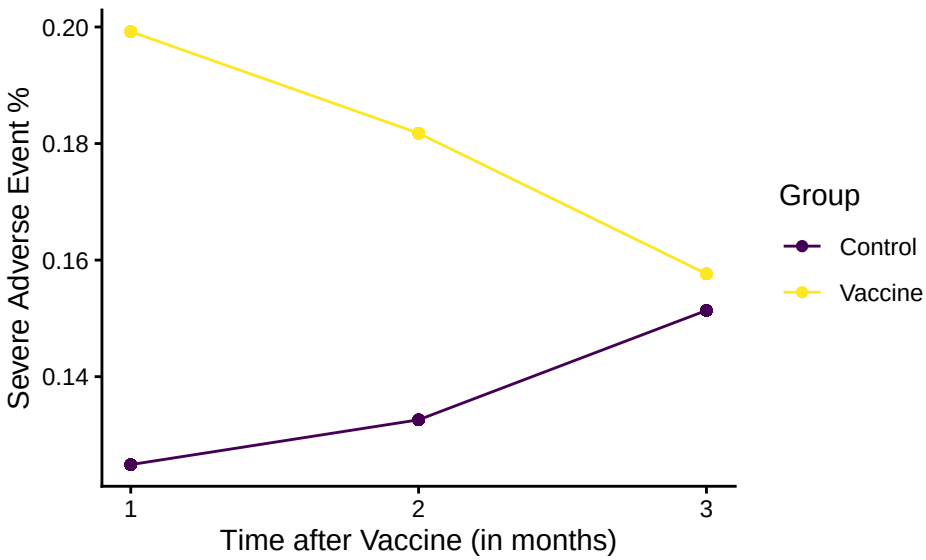
```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by group, time, site, and sex.
## i Output is grouped by group, time, and site.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(group, time, site, sex))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
sae_pct_by_gender_persite <- ggplot(aes(x = group_time, y = sae_pct, color = group),
       data = sae_pct) +
  geom_boxplot() +
  geom_jitter(alpha = 0.4) +
  stat_compare_means(comparisons = list(c("1_Control", "1_Vaccine"),
                                        c("2_Control", "2_Vaccine"),
                                        c("3_Control", "3_Vaccine")),
                     label = "p.adj.format",
                     p.adjust.method = "bonferroni") +
  labs(x = "Time after Vaccine (in months) + Tx Group", y = "Severe Adverse Event% (Per Site)", color =
  facet_wrap(~sex) +
  theme_classic()

ggsave("figures_tables/sae_pct_by_gender_persite.pdf", sae_pct_by_gender_persite, height = 5, width =11
#system("pdfcrop figures_tables/sae_pct_by_gender_persite.pdf figures_tables/sae_pct_by_gender_persite.
sae_pct_by_gender_persite
```



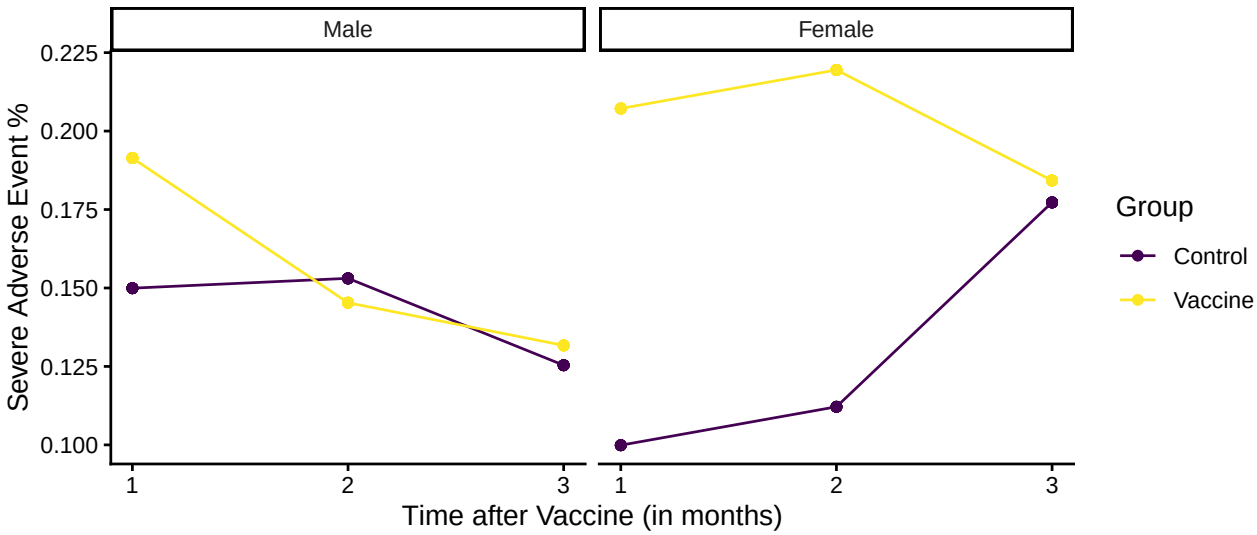

```
sae_pct <- combined_dataset |>
  drop_na() |>
  group_by(group, time) |>
```

```
  mutate(total = n()) |>
  ungroup() |>
  dplyr::filter(sae == "Yes") |>
  group_by(group, time) |>
  mutate (sae_pct = n()*100/total[1])

sae_pct_by_gender <- ggplot(aes(x = time, y = sae_pct, group = group, color = group),
       data = sae_pct) +
  geom_point() +
  scale_x_continuous(limits = c(1, 3), breaks = c(1, 2, 3))+
  geom_line() +
  labs(x = "Time after Vaccine (in months)", y = "Severe Adverse Event %", color = "Group") +
  theme_classic()

ggsave("figures_tables/sae_pct_by_gender.pdf", sae_pct_by_gender, height = 3, width =5)
#system("pdfcrop figures_tables/sae_pct_by_gender.pdf figures_tables/sae_pct_by_gender.pdf")
sae_pct_by_gender
```




```
sae_pct_by_group <- combined_dataset |>
  drop_na() |>
  group_by(sex, time, group) |>
  mutate(total = n()) |>
  ungroup() |>
  dplyr::filter(sae == "Yes") |>
  group_by(sex, time, group) |>
  mutate (sae_pct = n()*100/total[1])

sae_pct_by_gender_group <- ggplot(aes(x = time, y = sae_pct, group = group, color = group),
       data = sae_pct_by_group) +
  geom_point() +
  scale_x_continuous(limits = c(1, 3), breaks = c(1, 2, 3))+
  geom_line() +
  facet_wrap(~sex) +
  labs(x = "Time after Vaccine (in months)", y = "Severe Adverse Event %", color = "Group") +
  theme_classic()
```

```
ggsave("figures_tables/sae_pct_by_gender_group.pdf", sae_pct_by_gender_group, height = 3, width =7)
#system("pdfcrop figures_tables/sae_pct_by_gender_group.pdf figures_tables/sae_pct_by_gender_group.pdf"
sae_pct_by_gender_group
```



Modeling: GLMM to model for odds of SAE at each timepoint

```
combined_dataset <- combined_dataset |>
  mutate(time = as.factor(time))
glmm1 <- glmer(sae ~ group*time + group + sex + age + (1 | id),
               family = binomial, data = combined_dataset, nAGQ = 0)
glmm2 <- glmer(sae ~ group + time + group + sex + age + (1 | id),
               family = binomial, data = combined_dataset, nAGQ = 0)
anova(glmm1, glmm2)
```

```
## Data: combined_dataset
## Models:
## glmm2: sae ~ group + time + group + sex + age + (1 | id)
## glmm1: sae ~ group * time + group + sex + age + (1 | id)
##       npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## glmm2    7 2615.2 2682.5 -1300.6    2601.2
## glmm1    9 2617.8 2704.4 -1299.9    2599.8 1.3303  2     0.5142
```

Interaction term not needed

```
combined_dataset <- combined_dataset |>
  mutate(time = as.factor(time))
glmm3 <- glmer(sae ~ group*time + group + sex + age + (1 |site / id),
               family = binomial, data = combined_dataset, nAGQ = 0)
anova(glmm1, glmm3)
```

```
## Data: combined_dataset
## Models:
## glmm1: sae ~ group * time + group + sex + age + (1 | id)
## glmm3: sae ~ group * time + group + sex + age + (1 | site/id)
```

```
##       npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
## glmm1    9 2617.8 2704.4 -1299.9    2599.8
## glmm3   10 2618.6 2714.8 -1299.3    2598.6 1.259  1     0.2618
```

read on correlation structure, addin two separate random intercept etc.